SRI (k-inflation): N02LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 2$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

For k-inflation, $c_t = 1$. The scalar amplitude includes transport from the acoustic reference to the common tensor reference. The sound speed is $c_{s*} > 0$, with

$$\alpha_n^{c_s} = \frac{d^{n+1}\ln c_s}{d(\ln k)^{n+1}}.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*} c_{s*}} \left[\sum_{d=0}^{2} F_{s,d} + O(\mathrm{SR}^3)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$\begin{aligned} F_{s,1} = {} & 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1 \\ & - 1.2703628454615\alpha_0^{c_s} - 1.0\alpha_0^{c_s}\log(c_s) \\ & - 1.0\epsilon_2\log(c_s) - 2.0\epsilon_1\log(c_s) \end{aligned}$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & - 0.39567736285244\alpha_1^{c_s} - 0.22975141576082\left(\alpha_0^{c_s}\right)^2 \\ & + 0.5\left(\alpha_0^{c_s}\right)^2\log(c_s)^2 + 0.5\epsilon_2^2\log(c_s)^2 \\ & + 2.0\epsilon_1^2\log(c_s)^2 + 0.27036284546148\left(\alpha_0^{c_s}\right)^2\log(c_s) \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 1.0809943369567\alpha_0^{c_s}\epsilon_1 \\ & + 0.14504832807052\epsilon_2\epsilon_3 - 0.5\alpha_1^{c_s}\log(c_s)^2 \\ & - 0.91854861815409\epsilon_1^2\log(c_s) - 0.72986567698311\alpha_0^{c_s}\epsilon_2 \\ & - 1.2703628454615\alpha_1^{c_s}\log(c_s) - 0.72963715453852\epsilon_2^2\log(c_s) \\ & + 1.0\alpha_0^{c_s}\epsilon_2\log(c_s)^2 + 1.0\epsilon_1\epsilon_2\log(c_s)^2 \\ & + 2.0\alpha_0^{c_s}\epsilon_1\log(c_s)^2 + 1.0814513818459\alpha_0^{c_s}\epsilon_1\log(c_s) \\ & + 0.72963715453852\epsilon_2\epsilon_3\log(c_s) - 0.5\epsilon_2\epsilon_3\log(c_s)^2 \\ & - 0.45927430907704\alpha_0^{c_s}\epsilon_2\log(c_s) \\ & - 2.459274309077\epsilon_1\epsilon_2\log(c_s) \end{aligned}$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{2} F_{t,d} + O(\mathrm{SR}^3)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

Relative order 2.

$$F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2$$

2 Consistency relations

2.1 $n = 0$

For $p = 2$, the absolute cutoff is $W = 3$:

$$\alpha_0^t = \sum_{d=1}^{3} R_{0,d} + O(\mathrm{SR}^4).$$

Absolute order 1.

$$R_{0,1} = -\frac{0.125r}{c_s}$$

Absolute order 2.

$$\begin{aligned} R_{0,2} = {} & \frac{0.0078125r^2}{c_s^2} + \frac{0.125\alpha_0^s r}{c_s} \\ & + \frac{0.28379535568268\alpha_0^{c_s} r}{c_s} - \frac{0.125\alpha_0^s r \log(c_s)}{c_s} \end{aligned}$$

Absolute order 3.

$$\begin{aligned} R_{0,3} = {} & -\frac{0.0030712246746565r^3}{c_s^3} + \frac{0.125\alpha_1^s r}{c_s} \\ & + \frac{0.33325502603924\alpha_1^{c_s} r}{c_s} - \frac{0.10280837917801\left(\alpha_0^s\right)^2 r}{c_s} \\ & - \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^2 r}{c_s} - \frac{0.041327094794504\alpha_0^s r^2}{c_s^2} \\ & - \frac{0.061176514254839\alpha_0^{c_s} r^2}{c_s^2} + \frac{0.015625\alpha_0^s r^2 \log(c_s)}{c_s^2} \\ & + \frac{0.0625\alpha_1^s r \log(c_s)^2}{c_s} \\ & + \frac{0.125\left(\alpha_0^s\right)^2 r \log(c_s)}{c_s} \\ & - \frac{0.0625\left(\alpha_0^s\right)^2 r \log(c_s)^2}{c_s} \\ & - \frac{0.125\alpha_1^s r \log(c_s)}{c_s} - \frac{0.46478233218736\alpha_0^{c_s}\alpha_0^s r}{c_s} \\ & + \frac{0.28379535568268\alpha_0^{c_s}\alpha_0^s r \log(c_s)}{c_s} \end{aligned}$$

2.2 $n = 1$

For $p = 2$, the absolute cutoff is $W = 4$:

$$\alpha_1^t = \sum_{d=2}^{4} R_{1,d} + O(\mathrm{SR}^5).$$

Absolute order 2.

$$R_{1,2} = \frac{0.015625r^2}{c_s^2} + \frac{0.125\alpha_0^{c_s} r}{c_s} + \frac{0.125\alpha_0^s r}{c_s}$$

Absolute order 3.

$$\begin{aligned} R_{1,3} = & -\frac{0.0029296875r^3}{c_s^3} + \frac{0.125\alpha_1^s r}{c_s} \\ & + \frac{0.28379535568268\alpha_1^{c_s} r}{c_s} - \frac{0.125\left(\alpha_0^s\right)^2 r}{c_s} \\ & - \frac{0.046875\alpha_0^s r^2}{c_s^2} - \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2 r}{c_s} \\ & - \frac{0.086573838920671\alpha_0^{c_s} r^2}{c_s^2} + \frac{0.03125\alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\ & + \frac{0.125\left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\ & - \frac{0.125\alpha_1^s r \log\left(c_s\right)}{c_s} - \frac{0.53379535568268\alpha_0^{c_s}\alpha_0^s r}{c_s} \\ & + \frac{0.125\alpha_0^{c_s}\alpha_0^s r \log\left(c_s\right)}{c_s} \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{1,4} = & \frac{0.0016576826498282r^4}{c_s^4} + \frac{0.125\alpha_2^s r}{c_s} \\ & + \frac{0.12398128438351\left(\alpha_0^s\right)^2 r^2}{c_s^2} + \frac{0.10280837917801\left(\alpha_0^s\right)^3 r}{c_s} \\ & + \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^3 r}{c_s} \\ & + \frac{0.25495658892655\left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.038806320518347\alpha_0^{c_s} r^3}{c_s^3} \\ & + \frac{0.33325502603924\alpha_2^{c_s} r}{c_s} + \frac{0.027641022071908\alpha_0^s r^3}{c_s^3} \\ & - \frac{0.14449027076465\alpha_1^{c_s} r^2}{c_s^2} - \frac{0.072577094794504\alpha_1^s r^2}{c_s^2} \\ & + \frac{0.03125\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\ & + \frac{0.0625\alpha_2^s r \log\left(c_s\right)^2}{c_s} \\ & + \frac{0.0625\left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\ & + \frac{0.046875\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\ & + \frac{0.69259071136537\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r}{c_s} \\ & + \frac{0.4077766400662\alpha_0^{c_s}\alpha_0^s r^2}{c_s^2} \\ & + \frac{0.95683271390928\left(\alpha_0^{c_s}\right)^2\alpha_0^s r}{c_s} \\ & - \frac{0.015625\alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \end{aligned}$$

$$
\begin{aligned}
&- \frac{0.125\alpha_2^s r \log(c_s)}{c_s} \\
&- \frac{0.125\left(\alpha_0^s\right)^3 r \log(c_s)}{c_s} \\
&- \frac{0.09375\left(\alpha_0^s\right)^2 r^2 \log(c_s)}{c_s^2} \\
&- \frac{0.0087890625\alpha_0^s r^3 \log(c_s)}{c_s^3} - \frac{0.33061675835603\alpha_0^s \alpha_1^s r}{c_s} \\
&- \frac{0.71478233218736\alpha_0^{c_s} \alpha_1^s r}{c_s} - \frac{0.74976507811772\alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.79803735822659\alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r \log(c_s)^2}{c_s} \\
&+ \frac{0.375\alpha_0^s \alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{0.28379535568268\alpha_0^s \alpha_1^{c_s} r \log(c_s)}{c_s} \\
&+ \frac{0.53379535568268\alpha_0^{c_s} \alpha_1^s r \log(c_s)}{c_s} \\
&- \frac{0.0625\alpha_0^{c_s} \alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{0.1875\alpha_0^s \alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log(c_s)}{c_s} \\
&- \frac{0.17314767784134\alpha_0^{c_s} \alpha_0^s r^2 \log(c_s)}{c_s^2} \\
&- \frac{0.53379535568268\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r \log(c_s)}{c_s}
\end{aligned}
$$

2.3 $n = 2$

For $p = 2$, the absolute cutoff is $W = 5$:

$$\alpha_2^t = \sum_{d=3}^{5} R_{2,d} + O(\mathrm{SR}^6).$$

Absolute order 3.

$$
\begin{aligned}
R_{2,3} = &- \frac{0.00390625 r^3}{c_s^3} + \frac{0.125\alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.125\alpha_1^s r}{c_s} - \frac{0.125\left(\alpha_0^{c_s}\right)^2 r}{c_s} \\
&- \frac{0.125\left(\alpha_0^s\right)^2 r}{c_s} - \frac{0.046875\alpha_0^{c_s} r^2}{c_s^2} \\
&- \frac{0.046875\alpha_0^s r^2}{c_s^2} - \frac{0.25\alpha_0^{c_s} \alpha_0^s r}{c_s}
\end{aligned}
$$

Absolute order 4.

$$
\begin{aligned}
R_{2,4} = &\frac{0.0013427734375 r^4}{c_s^4} + \frac{0.125\alpha_2^s r}{c_s} \\
&+ \frac{0.125\left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.125\left(\alpha_0^s\right)^2 r^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.025390625\alpha_0^s r^3}{c_s^3} + \frac{0.28379535568268\alpha_2^{c_s} r}{c_s} \\
&+ \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^3 r}{c_s} \\
&+ \frac{0.24409651676201\left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.040277689595252\alpha_0^{c_s} r^3}{c_s^3} \\
&- \frac{0.0625\alpha_1^s r^2}{c_s^2} - \frac{0.12204825838101\alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.046875\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\
&+ \frac{0.78379535568268\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r}{c_s} \\
&+ \frac{0.41597151676201\alpha_0^{c_s}\alpha_0^s r^2}{c_s^2} - \frac{0.125\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.125\left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.01171875\alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.09375\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} - \frac{0.375\alpha_0^s\alpha_1^s r}{c_s} \\
&- \frac{0.81759071136537\alpha_0^s\alpha_1^{c_s} r}{c_s} - \frac{0.78379535568268\alpha_0^{c_s}\alpha_1^s r}{c_s} \\
&- \frac{0.85138606704805\alpha_0^{c_s}\alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.125\alpha_0^s\alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.25\alpha_0^{c_s}\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_0^s\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.125\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.25\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.09375\alpha_0^{c_s}\alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{2,5} = &- \frac{0.00099348164677838 r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s} \\
&+ \frac{0.050668108270534\alpha_1^s r^3}{c_s^3} + \frac{0.33325502603924\alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.087560258989506\alpha_1^{c_s} r^3}{c_s^3} - \frac{0.24566880066715\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.13499234851023\left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.28928966356152\left(\alpha_0^s\right)^3 r^2}{c_s^2} - \frac{0.020502543360439\alpha_0^s r^4}{c_s^4}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.10280837917801\left(\alpha_0^s\right)^4 r}{c_s} - \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^4 r}{c_s} \\
&- \frac{0.027325781299929\alpha_0^{c_s} r^4}{c_s^4} - \frac{0.64251673826997\left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
&- \frac{0.33061675835603\left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.088202094794504\alpha_2^s r^2}{c_s^2} \\
&- \frac{0.18614714901955\alpha_2^{c_s} r^2}{c_s^2} - \frac{0.74976507811772\left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{0.0625\alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} + \frac{0.0625\alpha_3^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.125\left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.25\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.005859375\alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.375\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.00537109375\alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.076171875\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{2.2054104017793\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.490628069592\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.0148110363091\alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.63904189589007\left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.3745301562354\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.74025304063896\alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.87364314908906\alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.46569385315053\alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&- \frac{0.0625\left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0234375\alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.09375\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.1875\left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.017578125\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.025390625\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.92039909054338\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1312923842658\alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.4488830956312\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{2.1832187809573\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.083020104157\alpha_0^{c_s}\alpha_2^{c_s} r}{c_s} \\
&- \frac{1.4169873656614\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{1.512819690414\alpha_1^{c_s}\alpha_1^s r}{c_s} - \frac{0.38983219984103\alpha_0^{c_s}\alpha_0^s r^3}{c_s^3} \\
&- \frac{0.45561675835603\alpha_0^s\alpha_2^s r}{c_s} \\
&- \frac{1.8893109571319\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{0.96478233218736\alpha_0^{c_s}\alpha_2^s r}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.5\alpha_0^s\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.046875\alpha_0^{c_s}\alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.375\left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.140625\alpha_0^s\alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.28379535568268\alpha_0^s\alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.81759071136537\alpha_1^{c_s}\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.78379535568268\alpha_0^{c_s}\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.78379535568268\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.48819303352403\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.83194303352403\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.12083306878576\alpha_0^{c_s}\alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{2.8055805132741\alpha_0^{c_s}\alpha_0^s\alpha_1^s r}{c_s} \\
&+ \frac{3.7452632198456\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.0625\alpha_1^{c_s}\alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.0625\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125\alpha_0^{c_s}\alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.25\alpha_0^s\alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.09375\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.375\alpha_0^s\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.81759071136537\left(\alpha_0^s\right)^2\alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.24409651676201\alpha_0^s\alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.41597151676201\alpha_0^{c_s}\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.375\alpha_0^{c_s}\alpha_0^s\alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.85138606704805\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{2.3513860670481\alpha_0^{c_s}\alpha_0^s\alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

2.4 $n = 3$

For $p = 2$, the absolute cutoff is $W = 6$:

$$\alpha_3^t = \sum_{d=4}^{6} R_{3,d} + O(\mathrm{SR}^7).$$

Absolute order 4.

$$
\begin{aligned}
R_{3,4} = &\frac{0.00146484375 r^4}{c_s^4} + \frac{0.125\alpha_2^{c_s} r}{c_s} \\
&+ \frac{0.125\alpha_2^s r}{c_s} + \frac{0.125\left(\alpha_0^{c_s}\right)^3 r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.125\left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.0234375\alpha_0^{c_s} r^3}{c_s^3} \\
&+ \frac{0.0234375\alpha_0^s r^3}{c_s^3} + \frac{0.109375\left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{0.109375\left(\alpha_0^s\right)^2 r^2}{c_s^2} - \frac{0.0625\alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{0.0625\alpha_1^s r^2}{c_s^2} + \frac{0.375\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r}{c_s} \\
&+ \frac{0.375\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} + \frac{0.21875\alpha_0^{c_s}\alpha_0^s r^2}{c_s^2} \\
&- \frac{0.375\alpha_0^{c_s}\alpha_1^{c_s} r}{c_s} - \frac{0.375\alpha_0^{c_s}\alpha_1^s r}{c_s} \\
&- \frac{0.375\alpha_0^s\alpha_1^{c_s} r}{c_s} - \frac{0.375\alpha_0^s\alpha_1^s r}{c_s}
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{3,5} = &- \frac{0.000762939453125 r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s} \\
&+ \frac{0.0419921875\alpha_1^s r^3}{c_s^3} + \frac{0.28379535568268\alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.071766316690503\alpha_1^{c_s} r^3}{c_s^3} - \frac{0.125\left(\alpha_0^s\right)^4 r}{c_s} \\
&- \frac{0.375\left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.078125\alpha_2^s r^2}{c_s^2} \\
&- \frac{0.28125\left(\alpha_0^s\right)^3 r^2}{c_s^2} - \frac{0.01708984375\alpha_0^s r^4}{c_s^4} \\
&- \frac{0.1201171875\left(\alpha_0^s\right)^2 r^3}{c_s^3} - \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^4 r}{c_s} \\
&- \frac{0.15752267784134\alpha_2^{c_s} r^2}{c_s^2} - \frac{0.85138606704805\left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&- \frac{0.024533376047626\alpha_0^{c_s} r^4}{c_s^4} - \frac{0.5591418724447\left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
&- \frac{0.20943957507151\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&+ \frac{0.0625\alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.125\left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.005859375\alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.375\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.75\left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.21875\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.4375\alpha_0^s\alpha_1^s r^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.0703125\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.72129535568268\alpha_0^{c_s}\alpha_1^s r^2}{c_s^2} \\
&+ \frac{1.7027721340961\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.6013860670481\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.7778918724447\alpha_0^s\alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.87418722812738\alpha_0^{c_s}\alpha_1^{c_s} r^2}{c_s^2} - \frac{0.125\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.5\alpha_0^s\alpha_2^s r}{c_s} - \frac{0.0234375\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{1.1351814227307\alpha_0^{c_s}\alpha_2^{c_s} r}{c_s} - \frac{1.0337953556827\alpha_0^{c_s}\alpha_2^s r}{c_s} \\
&- \frac{1.0337953556827\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1013860670481\alpha_0^s\alpha_2^{c_s} r}{c_s} \\
&- \frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{1.6013860670481\alpha_1^{c_s}\alpha_1^s r}{c_s} \\
&- \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.6182837448894\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{1.3403918724447\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{0.35299426257151\alpha_0^{c_s}\alpha_0^s r^3}{c_s^3} \\
&+ \frac{0.125\alpha_0^s\alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.125\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.5\alpha_0^s\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_1^{c_s}\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.21875\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.4375\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.0703125\alpha_0^{c_s}\alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{3.1013860670481\alpha_0^{c_s}\alpha_0^s\alpha_1^s r}{c_s} \\
&+ \frac{3.6791582011442\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.125\alpha_0^s\alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.375\left(\alpha_0^s\right)^2\alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{0.375\left(\alpha_0^{c_s}\right)^2\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.21875\alpha_0^{c_s}\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.34375\alpha_0^s\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.375\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{1.125\alpha_0^{c_s}\alpha_0^s\alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{3,6} = &\frac{0.00069887861331449 r^6}{c_s^6} + \frac{0.125\alpha_4^s r}{c_s} \\
&+ \frac{1.065570626349\left(\alpha_0^{c_s}\right)^3 r^3}{c_s^3} \\
&+ \frac{1.4176370369568\left(\alpha_0^{c_s}\right)^4 r^2}{c_s^2} \\
&+ \frac{0.52264594794504\left(\alpha_1^s\right)^2 r^2}{c_s^2} + \frac{0.10280837917801\left(\alpha_0^s\right)^5 r}{c_s} \\
&+ \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^5 r}{c_s} \\
&+ \frac{0.24809448714071\left(\alpha_0^{c_s}\right)^2 r^4}{c_s^4} \\
&+ \frac{0.52433019631832\left(\alpha_0^s\right)^3 r^3}{c_s^3} + \frac{0.13631419746067\alpha_2^{c_s} r^3}{c_s^3} \\
&+ \frac{0.017257388611671\alpha_0^s r^5}{c_s^5} + \frac{1.1501885493287\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{0.07369519446916\alpha_2^s r^3}{c_s^3} + \frac{0.1495700174229\left(\alpha_0^s\right)^2 r^4}{c_s^4} \\
&+ \frac{0.61990642191755\left(\alpha_0^s\right)^4 r^2}{c_s^2} + \frac{0.33325502603924\alpha_4^{c_s} r}{c_s} \\
&+ \frac{0.022103438284604\alpha_0^{c_s} r^5}{c_s^5} - \frac{0.042328393296221\alpha_1^s r^4}{c_s^4} \\
&- \frac{0.066357117878927\alpha_1^{c_s} r^4}{c_s^4} - \frac{0.1038270947945\alpha_3^s r^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.22780402727446\alpha_3^{c_s} r^2}{c_s^2} + \frac{0.0625\alpha_4^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.01171875\alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.01171875\left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4} \\
&+ \frac{0.078125\alpha_3^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.21875\left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.171875\left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.10546875\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.01708984375\alpha_1^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{4.5746925879539\left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r}{c_s} \\
&+ \frac{1.1352294412117\alpha_0^{c_s} \alpha_2^s r^2}{c_s^2} \\
&+ \frac{2.6219204538578\left(\alpha_0^s\right)^2 \alpha_2^{c_s} r}{c_s} \\
&+ \frac{3.8874132271834\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r}{c_s} \\
&+ \frac{2.2571479479676\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&+ \frac{1.9409334773531\left(\alpha_0^{c_s}\right)^4 \alpha_0^s r}{c_s} \\
&+ \frac{3.5111972716302\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r}{c_s} \\
&+ \frac{8.2127818560984\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&+ \frac{4.4950282979633\alpha_0^s \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{1.6087005501362\alpha_0^s \left(\alpha_1^s\right)^2 r}{c_s} \\
&+ \frac{1.0946586542461\left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s} \\
&+ \frac{0.71467513753404\alpha_0^s \alpha_2^s r^2}{c_s^2} \\
&+ \frac{1.6030226512749\alpha_0^{c_s} \alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{0.40191656771049\alpha_0^{c_s} \alpha_0^s r^4}{c_s^4} \\
&+ \frac{1.1482074697214\alpha_0^{c_s} \left(\alpha_0^s\right)^4 r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{2.4575502603924\left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r}{c_s} \\
&+ \frac{1.4228234146217\alpha_0^s \alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{2.8877107655697\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3}{c_s^3} \\
&+ \frac{6.1460976449703\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2}{c_s^2} \\
&+ \frac{3.4988253905886\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{1.8602805292847\alpha_1^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{3.9539880896494\left(\alpha_0^{c_s}\right)^2 \alpha_2^s r}{c_s} \\
&+ \frac{3.9948526700024\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&- \frac{0.03125\alpha_3^s r^2 \log\left(c_s\right)^2}{c_s^2} - \frac{0.125\alpha_4^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.125\left(\alpha_0^s\right)^5 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0029296875\alpha_1^s r^4 \log\left(c_s\right)^2}{c_s^4} \\
&- \frac{0.4375\left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.5625\left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.068359375\left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)}{c_s^4} \\
&- \frac{0.0419921875\alpha_2^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.003814697265625\alpha_0^s r^5 \log\left(c_s\right)}{c_s^5} \\
&- \frac{0.3603515625\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.57385180946901\alpha_0^s \alpha_1^s r^3}{c_s^3} \\
&- \frac{2.4110271601353\left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{1.2147823321874\alpha_0^{c_s} \alpha_3^s r}{c_s} - \frac{1.4645474103051\alpha_0^s \alpha_3^{c_s} r}{c_s} \\
&- \frac{2.6441120746798\alpha_1^s \alpha_2^{c_s} r}{c_s} \\
&- \frac{4.5968842087759\left(\alpha_0^{c_s}\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{1.9544879813691\left(\alpha_0^s\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{4.4382967251885\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1.1168502750681\alpha_1^s\alpha_2^s r}{c_s} - \frac{1.1225357717973\alpha_0^{c_s}\alpha_1^{c_s} r^3}{c_s^3} \\
&- \frac{0.81325404477083\alpha_0^{c_s}\alpha_1^s r^3}{c_s^3} \\
&- \frac{1.0502754126021\left(\alpha_0^s\right)^3\alpha_1^s r}{c_s} \\
&- \frac{4.3255592153975\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2}{c_s^2} \\
&- \frac{2.2075502603924\left(\alpha_0^{c_s}\right)^3\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.58061675835603\alpha_0^s\alpha_3^s r}{c_s} - \frac{2.4776020226013\alpha_1^{c_s}\alpha_2^s r}{c_s} \\
&- \frac{2.5825502603924\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s} - \frac{1.4162751301962\alpha_0^{c_s}\alpha_3^{c_s} r}{c_s} \\
&- \frac{3.5073089371114\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{0.9442318016501\alpha_0^s\alpha_1^{c_s} r^3}{c_s^3} \\
&+ \frac{0.0625\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.625\alpha_0^s\alpha_3^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.625\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.25\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{1.25\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.109375\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.21875\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.4375\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.5625\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.234375\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$+\frac{0.9375\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.10546875\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.71875\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.3662109375\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.28379535568268\alpha_0^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.0337953556827\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.1013860670481\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.6013860670481\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.6013860670481\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.5557837448894\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{6.243371063951\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{1.6182837448894\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.21529895007151\alpha_0^s\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{4.7259796038175\alpha_0^{c_s}\alpha_0^s\alpha_2^s r}{c_s}$$
$$+\frac{0.35299426257151\alpha_0^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{7.299656342872\alpha_0^s\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{10.486494425184\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$-\frac{0.0625\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.0625\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.125\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.1875\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.1875\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.1875\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.3125\alpha_0^s\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$

$$
\begin{aligned}
&- \frac{0.625\alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.625\left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{1.25\left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.109375\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.03515625\alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{1.875\alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.59375\alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.10546875\alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.671875\left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^4 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.31504535568268\alpha_0^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.72129535568268\alpha_0^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.85138606704805\alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{1.0337953556827\alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&- \frac{1.1013860670481\left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.098133504190503\alpha_0^{c_s} \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&- \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&- \frac{1.1182837448894\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.1013860670481\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.7778918724447\alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.2365674897788\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{2.6807837448894\alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.62831872521453 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{10.317828730023 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&- \frac{10.419919562718 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&- \frac{6.2653115524259 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&- \frac{1.0589827877145 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{11.728814544016 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r}{c_s} \\
&- \frac{8.7476479574518 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{6.9558196807943 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.1875 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.75 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.5625 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.7027721340961 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{6.2027721340961 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.7483744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{3.6791582011442 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{5.9291582011442 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{4.1233744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.1875 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.5625 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{1.125 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.65625 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{1.1351814227307 \alpha_0^{c_s} \alpha_0^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{4.1351814227307 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{3.6791582011442 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{4.8041582011442 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$